Docker usage

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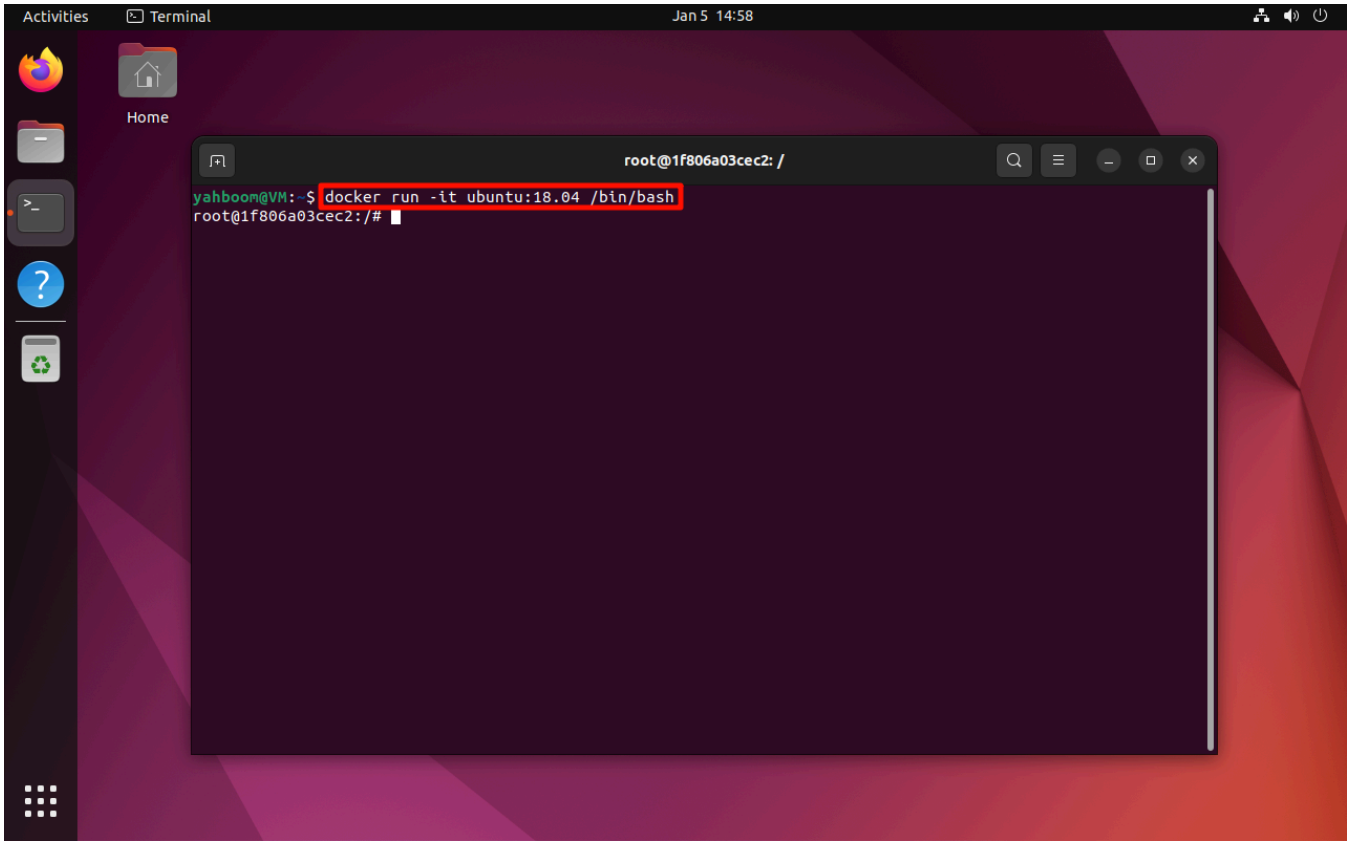
1. Start the image
2. ROS environment construction
    - 2.1. Update the system software
    - 2.2. Determine the language environment
        - 2.2.1. Verify the system environment
        - 2.2.2. Set the UTF-8 environment
    - 2.3. Set the software source
    - 2.4. Set the key
    - 2.5. Install the ROS1 desktop version
    - 2.6. Install the ROS1 dependencies
    - 2.7. rosdep initialization
    - 2.8. Set the environment variables
        - 2.8.1. Temporary setting
        - 2.8.2. Automatic setting (recommended)
3. Commit the image
4. Verify the image/ROS
    - 4.1. Start the image
    - 4.2. Verify the ROS environment

> **Historical demonstration:** This section shows how to build ROS 1 Melodic in an `ubuntu:18.04` container, only to help you understand the process of installing inside a container and committing the image. Ubuntu 18.04 and ROS 1 Melodic have both reached the end of their regular support, so this procedure is not an installation guide for the current Ubuntu 24.04 / ROS 2 Jazzy, and the repositories and dependencies may no longer be available.

1. Start the image

Start the `ubuntu:18.04` image in interactive mode:

```
docker run -it ubuntu:18.04 /bin/bash
```



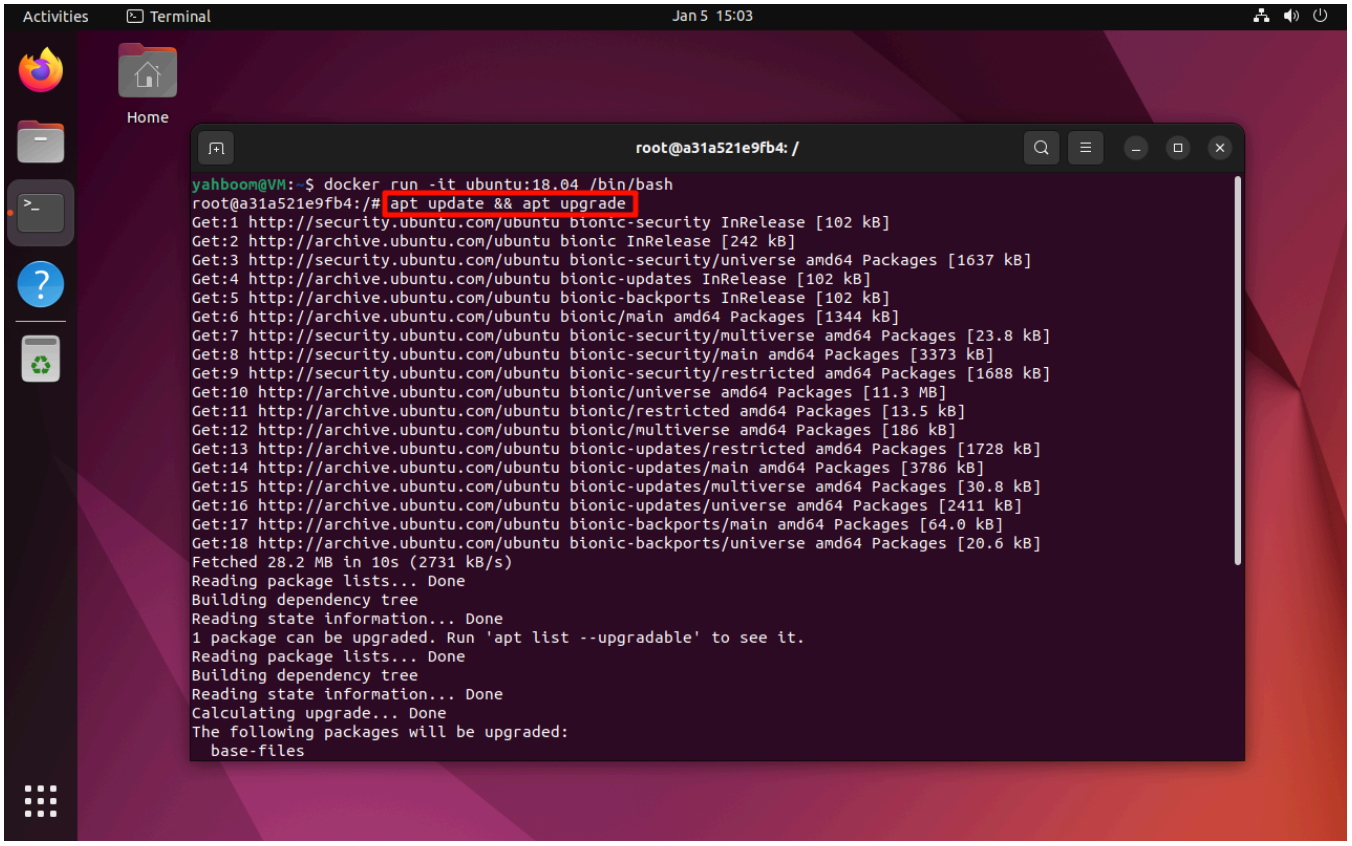
2. ROS environment construction

Administrator privileges are not needed inside Docker; if the host has the ROS environment installed, you need to add sudo to the commands.

2.1. Update the system software

To install the ROS1 environment, you generally need to make sure the system packages are up to date:

```
apt update && apt upgrade
```

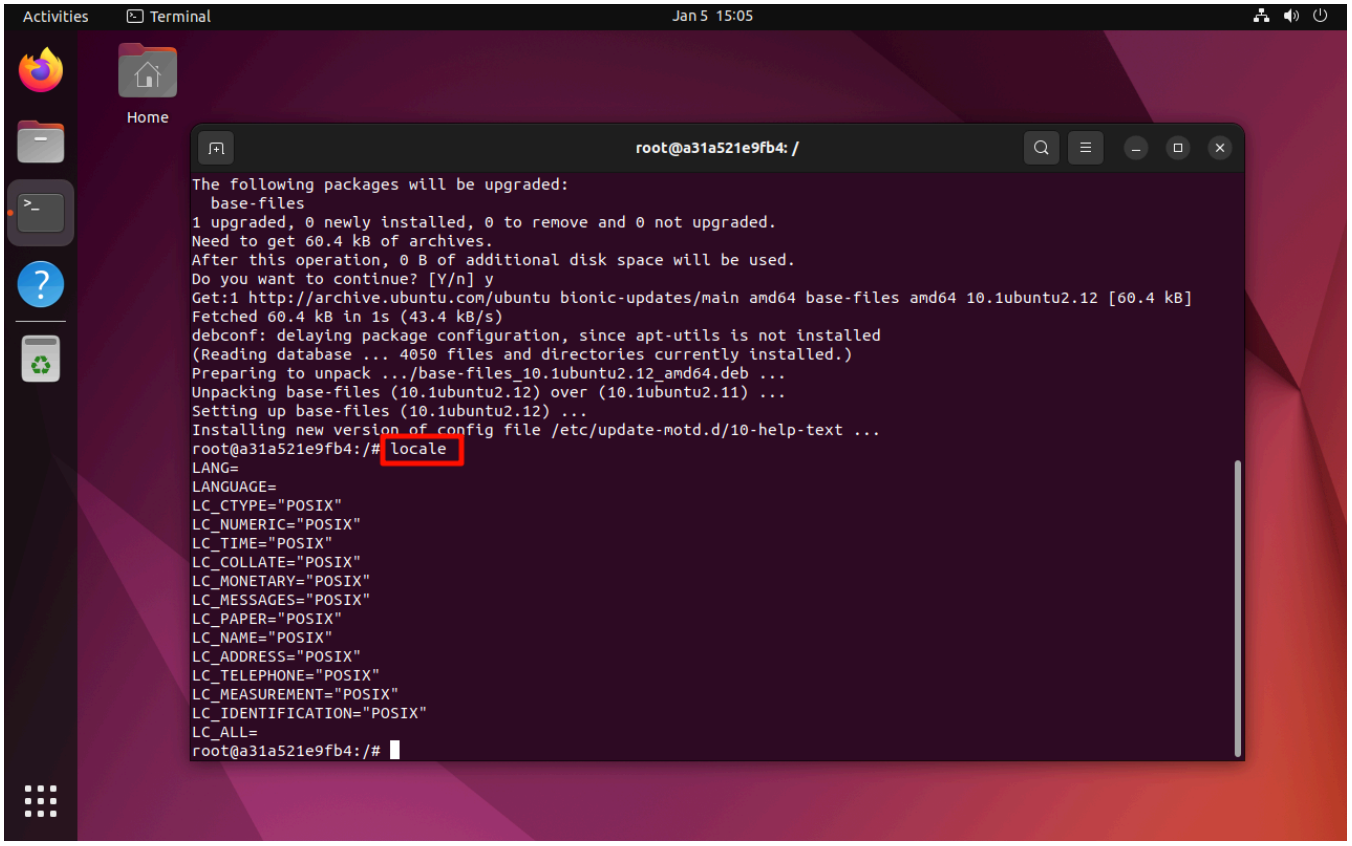


2.2. Determine the language environment

Make sure the system locale supports UTF-8!

2.2.1. Verify the system environment

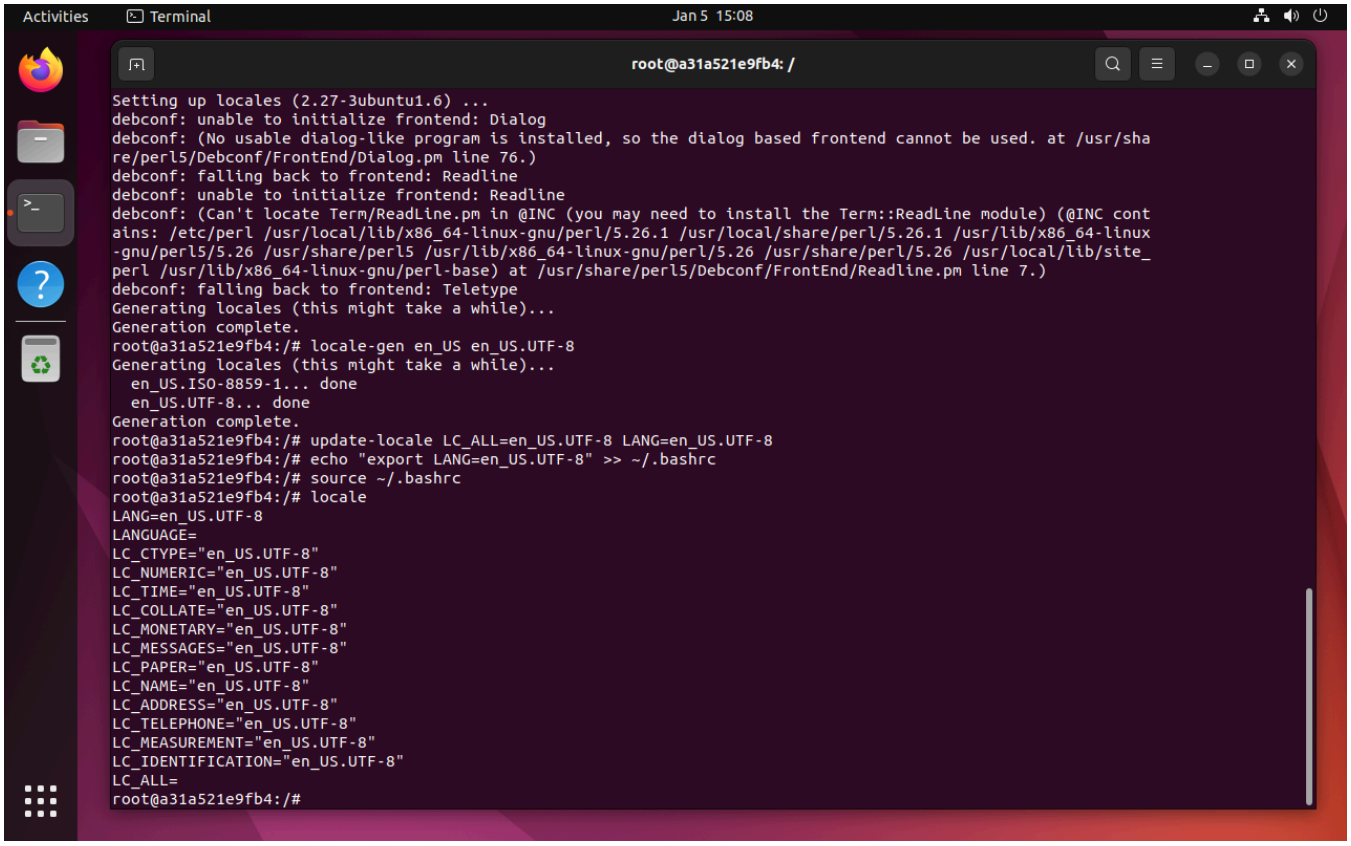
```
locale
```



2.2.2. Set the UTF-8 environment

If the system language does not support the UTF-8 environment, enter the following commands line by line in the terminal:

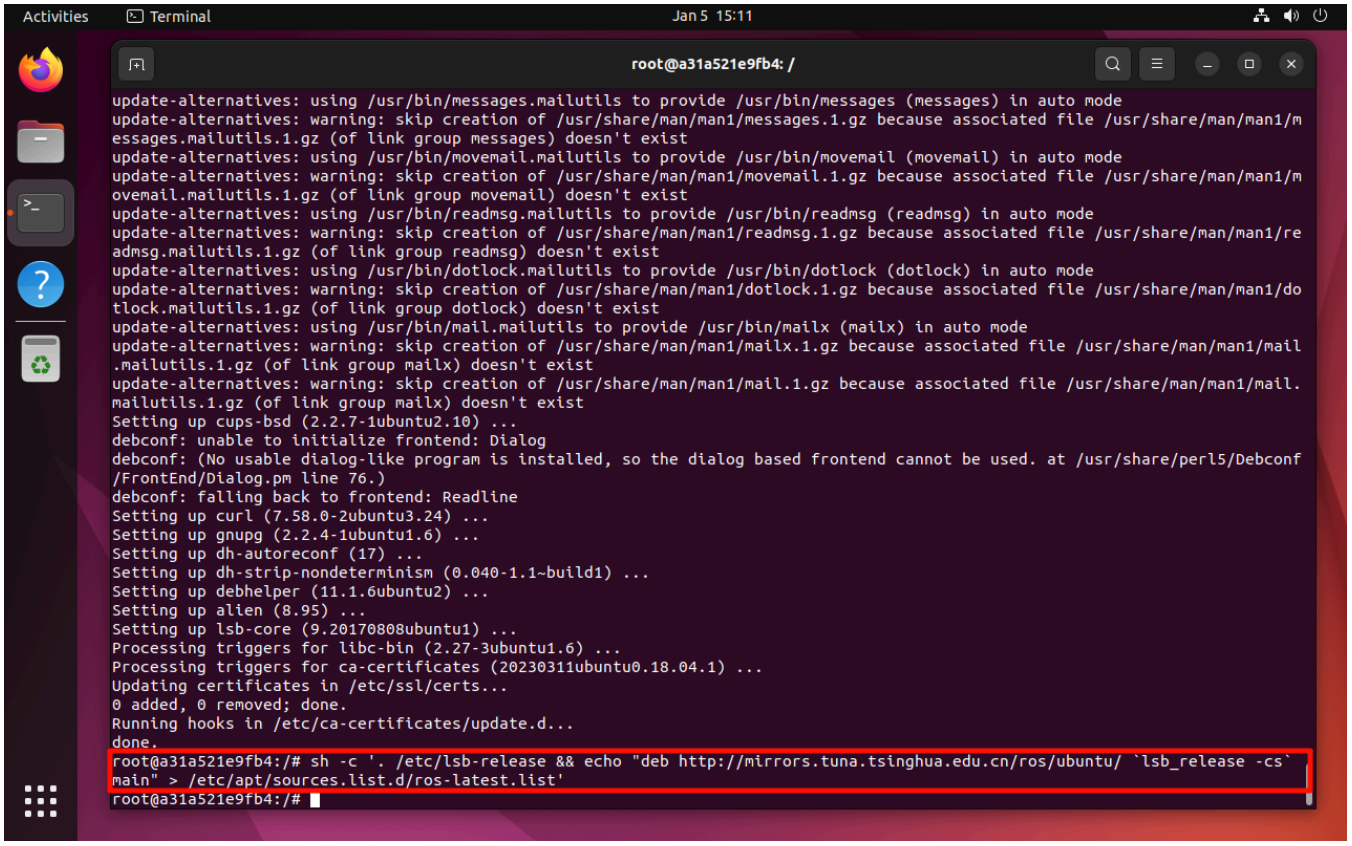
```
apt install locales
locale-gen en_US en_US.UTF-8
update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
echo "export LANG=en_US.UTF-8" >> ~/.bashrc
source ~/.bashrc
```



2.3. Set the software source

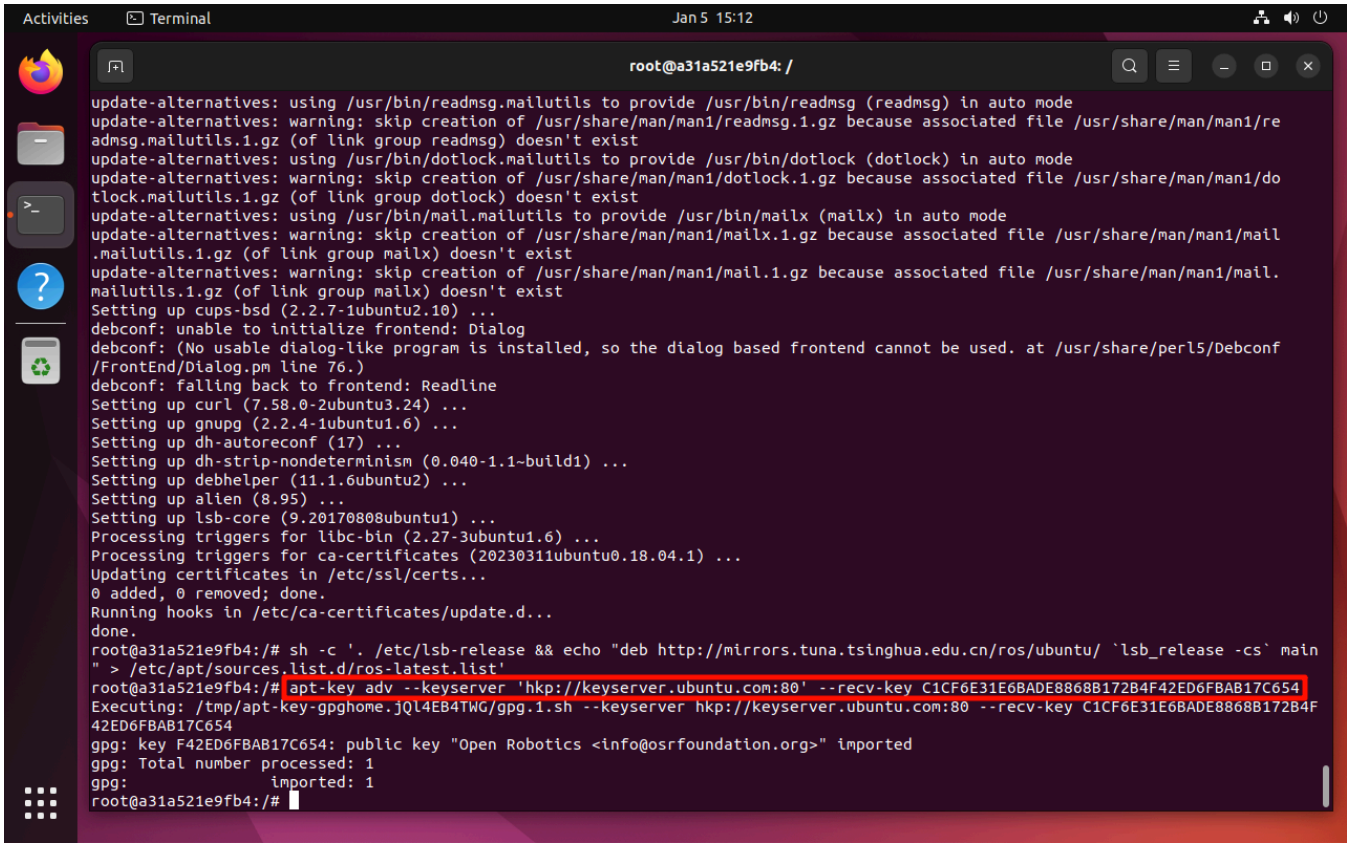
```
apt install lsb-core
```

```
sh -c '. /etc/lsb-release && echo "deb http://mirrors.tuna.tsinghua.edu.cn/ros/ubuntu/ `lsb_release -cs` main" > /etc/apt/sources.list.d/ros-latest.list'
```



2.4. Set the key

```
apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key
C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654
```

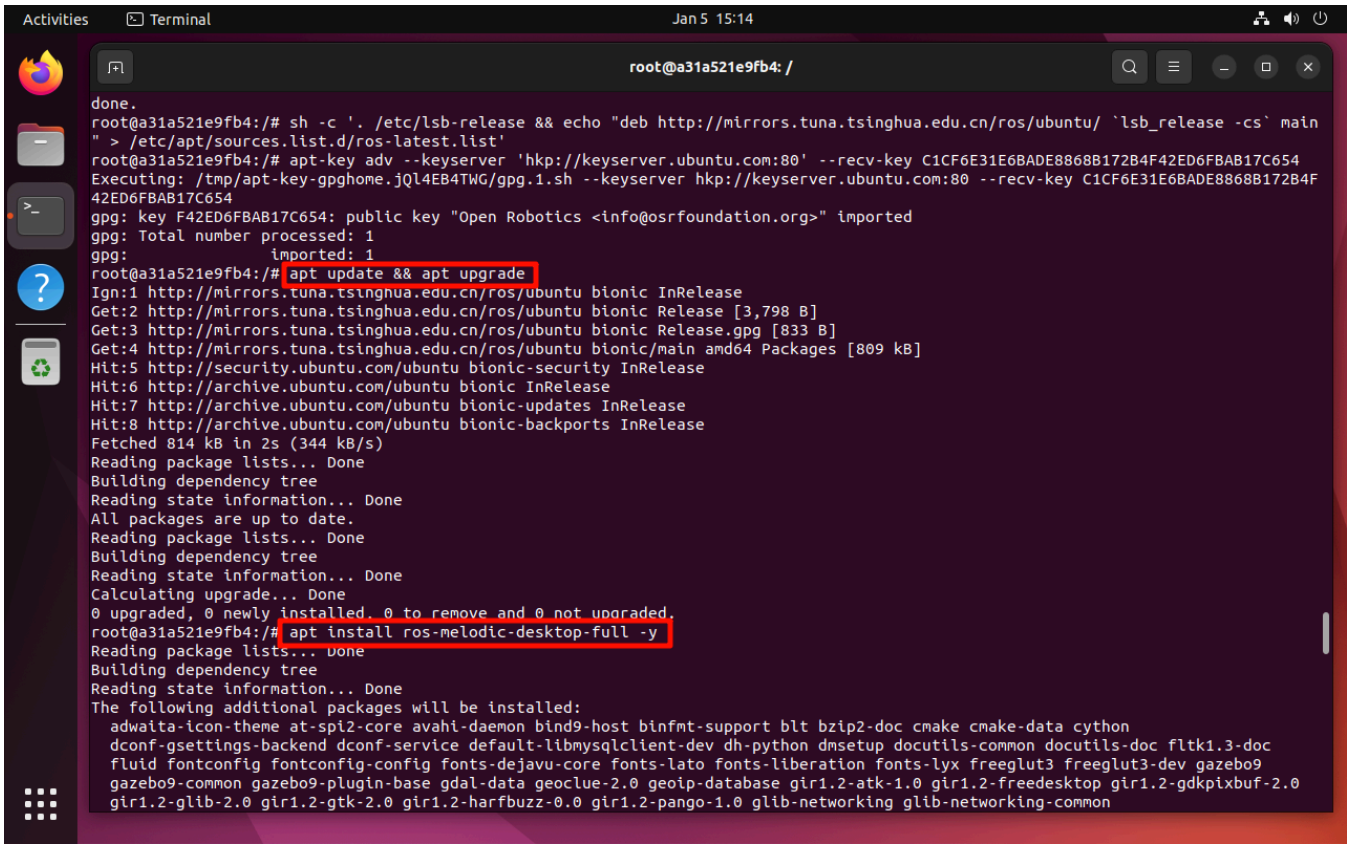


2.5. Install the ROS1 desktop version

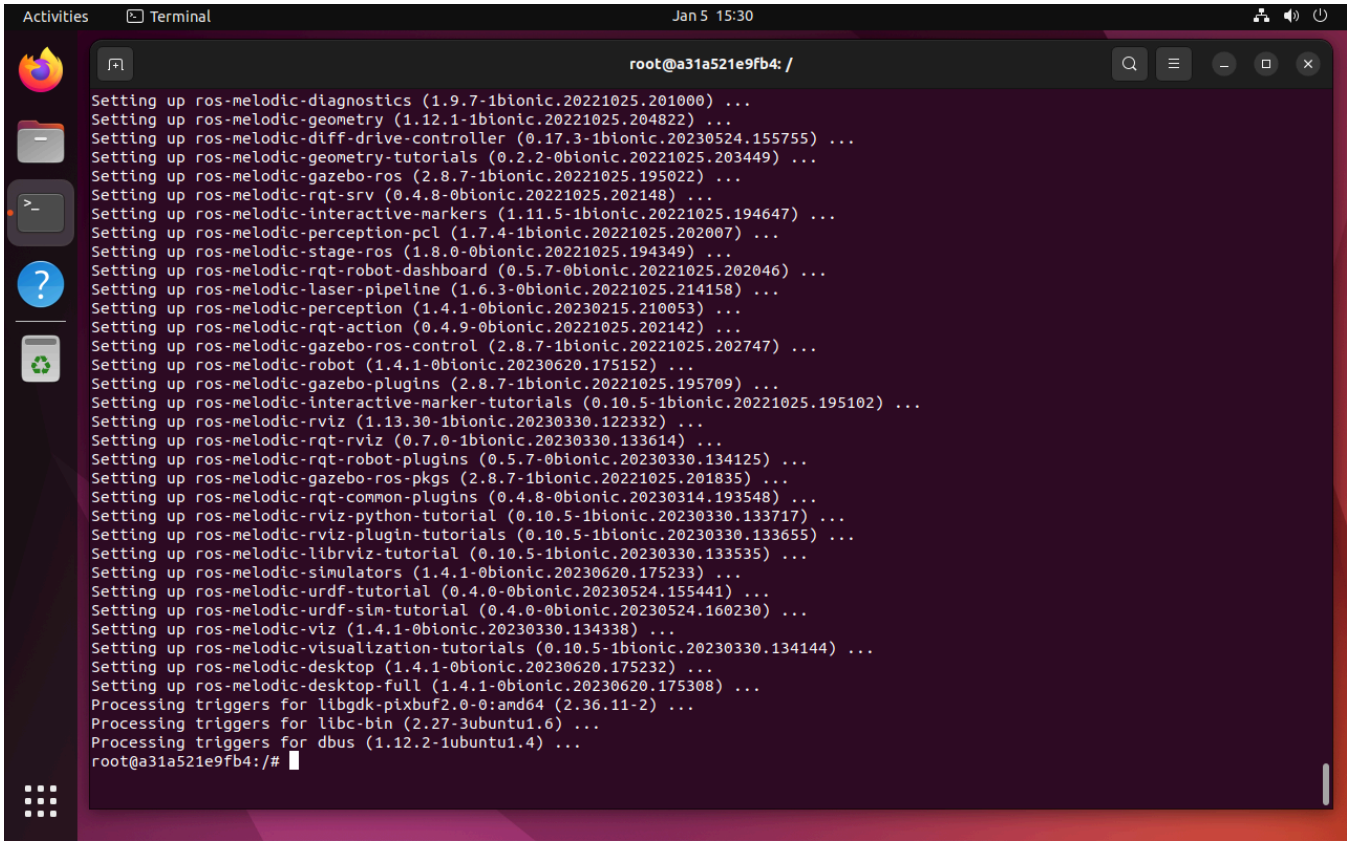
The ROS desktop version includes the ROS1 core features, common tools and libraries:

```
apt update && apt upgrade
```

```
apt install ros-melodic-desktop-full -y
```

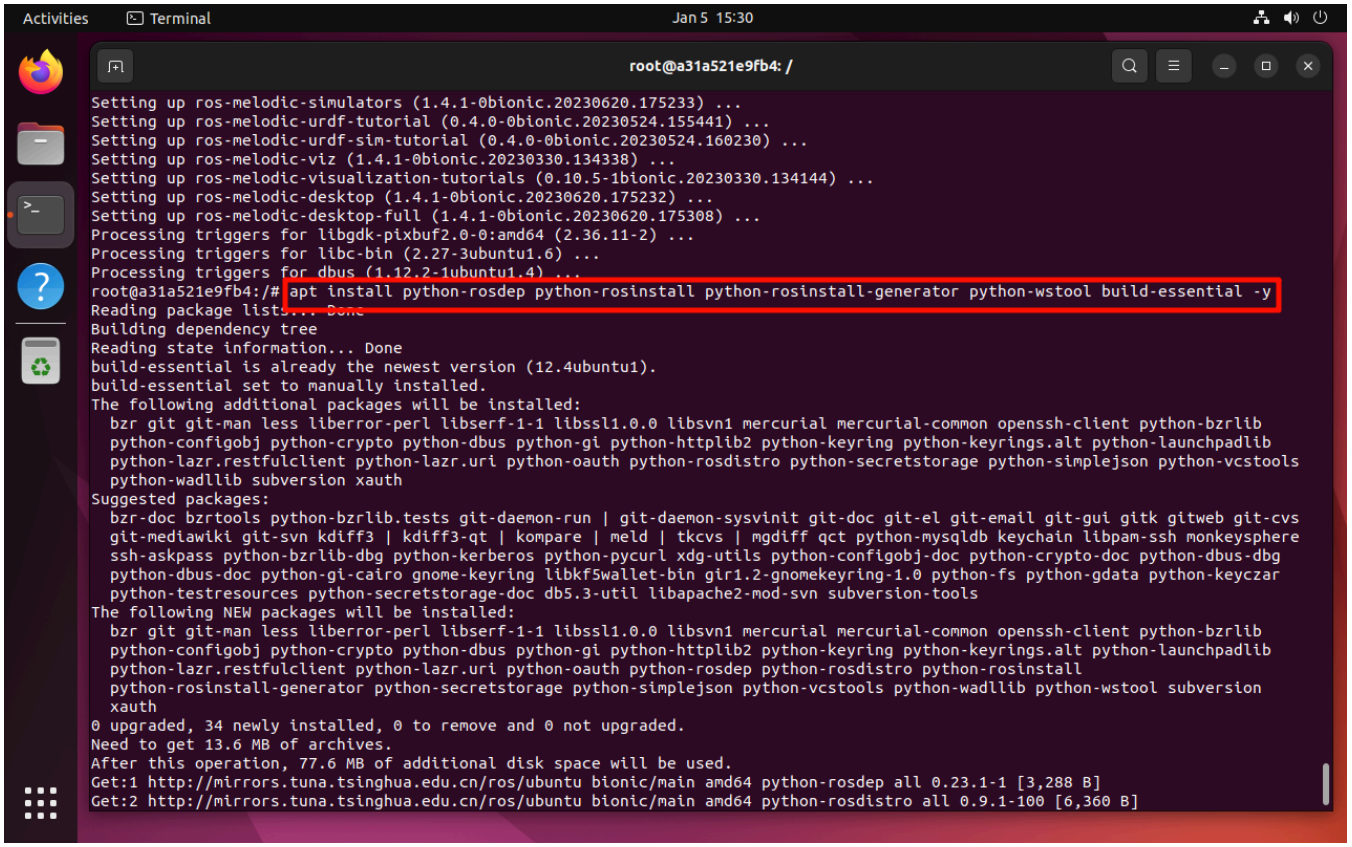


You need to select the region and city by yourself in the middle: this is not demonstrated here



2.6. Install the ROS1 dependencies

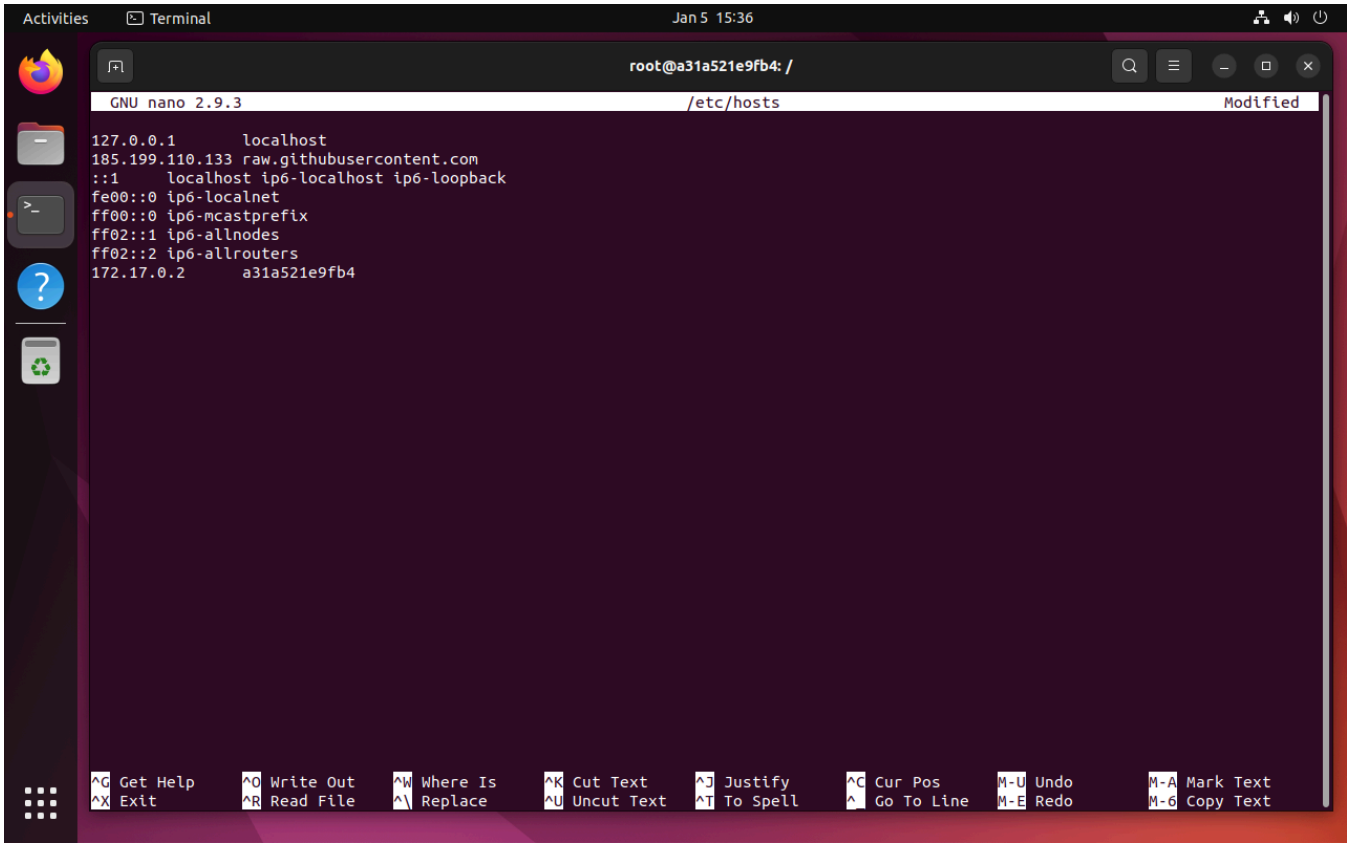
```
apt install python-rosdep python-rosinstall python-rosinstall-generator python-wstool build-essential -y
```



2.7. rosdep initialization

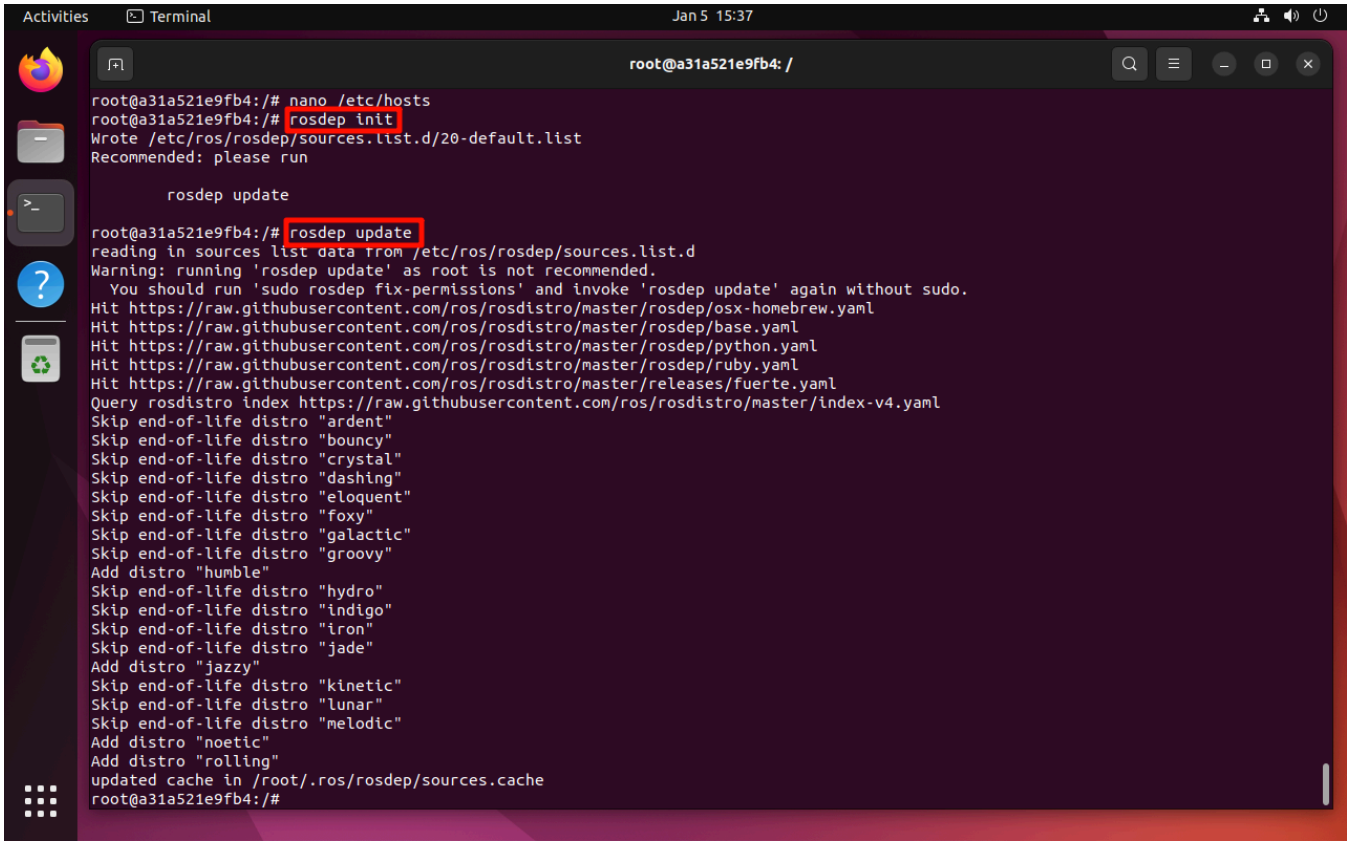
```
rosdep init
rosdep update
```

If `rosdep init` reports a download error, first check the container's DNS, network proxy and the HTTPS connectivity to `raw.githubusercontent.com`. Do not write a fixed IP address looked up from a third party into `/etc/hosts`; GitHub addresses may change, and a fixed resolution will cause subsequent connection errors.



Re-run:

```
rosdep init
rosdep update
```



2.8. Set the environment variables

2.8.1. Temporary setting

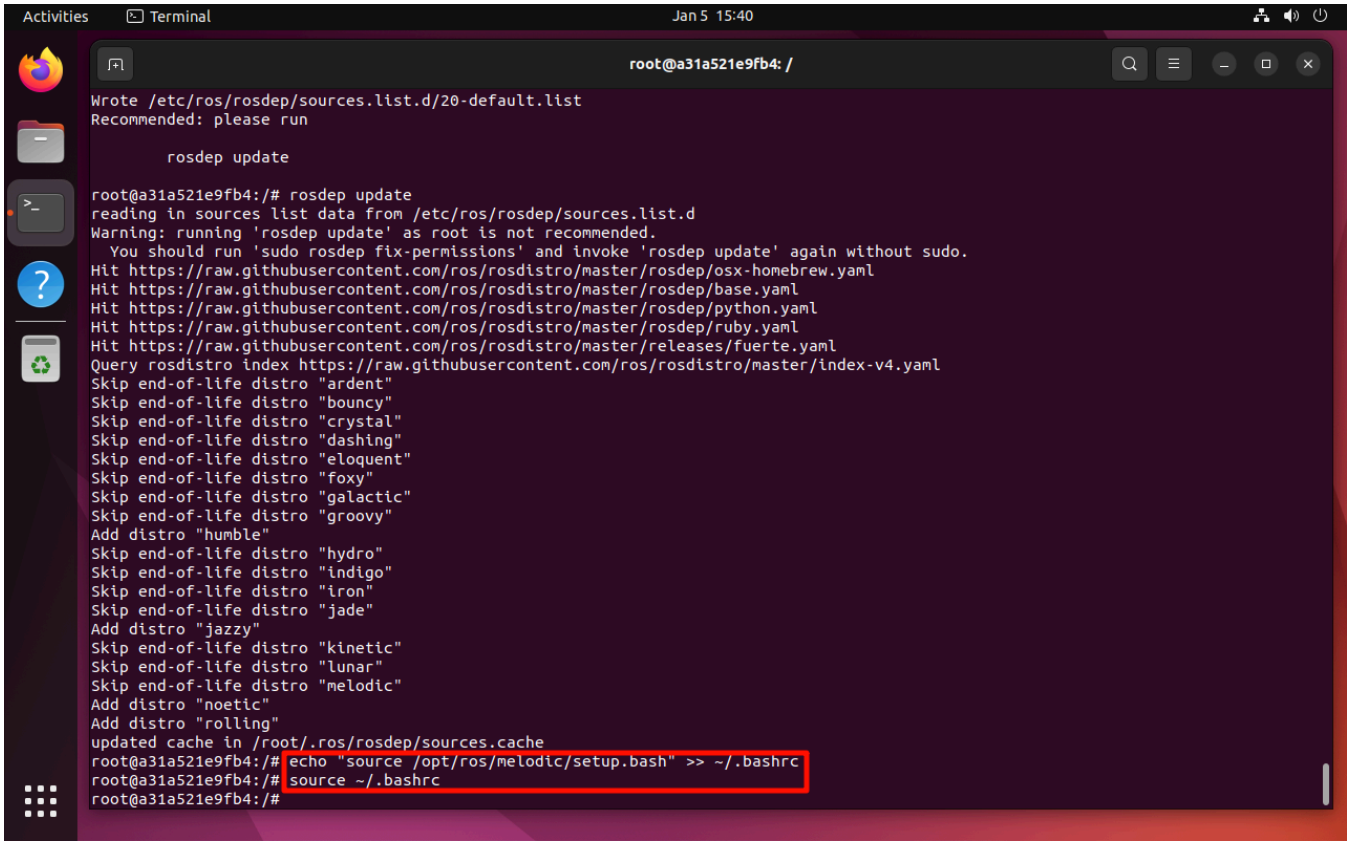
Refresh the environment variables of the current terminal:

```
source /opt/ros/melodic/setup.bash
```

2.8.2. Automatic setting (recommended)

Automatically refresh the environment variables when opening a terminal: opening a terminal will automatically refresh the contents of the .bashrc file

```
echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

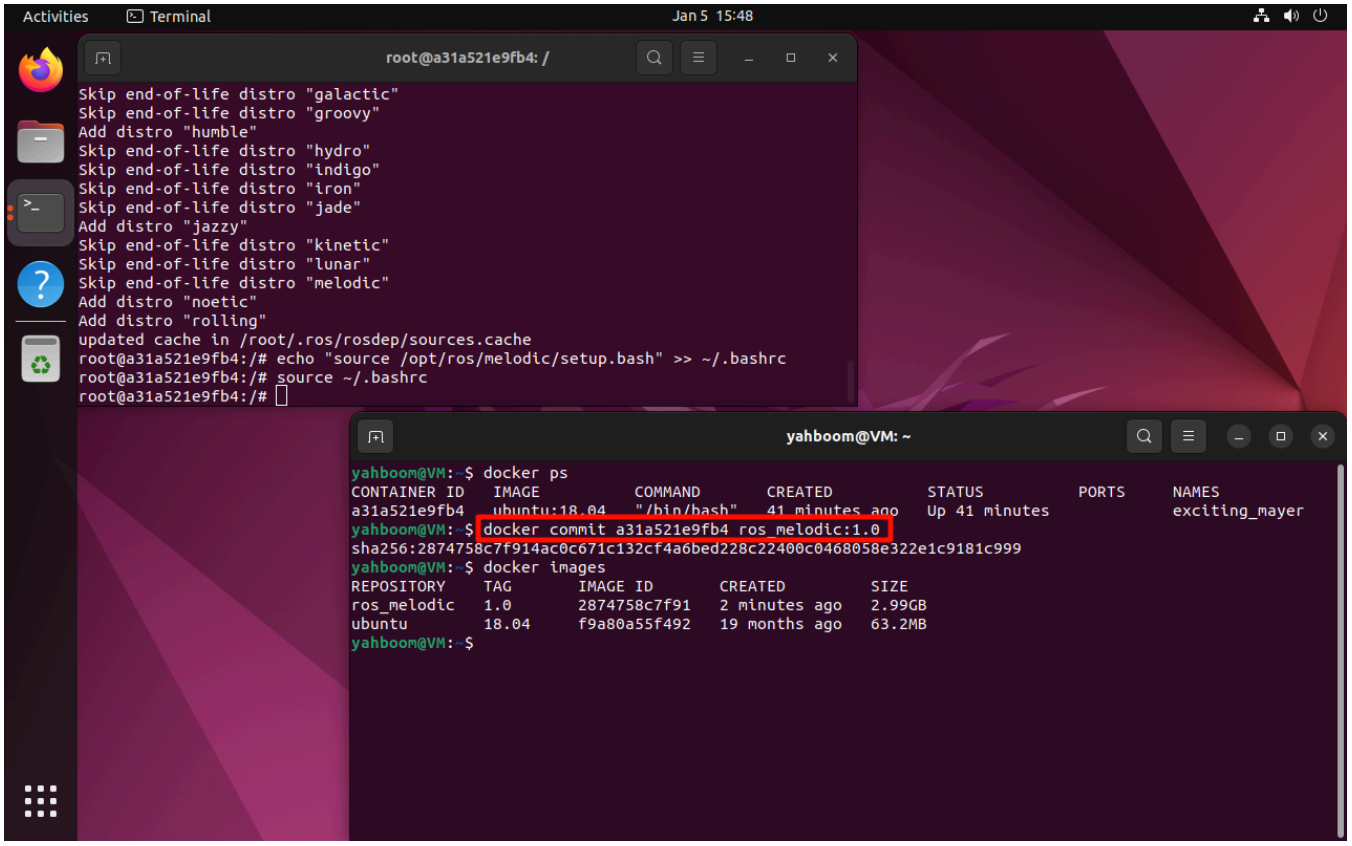


3. Commit the image

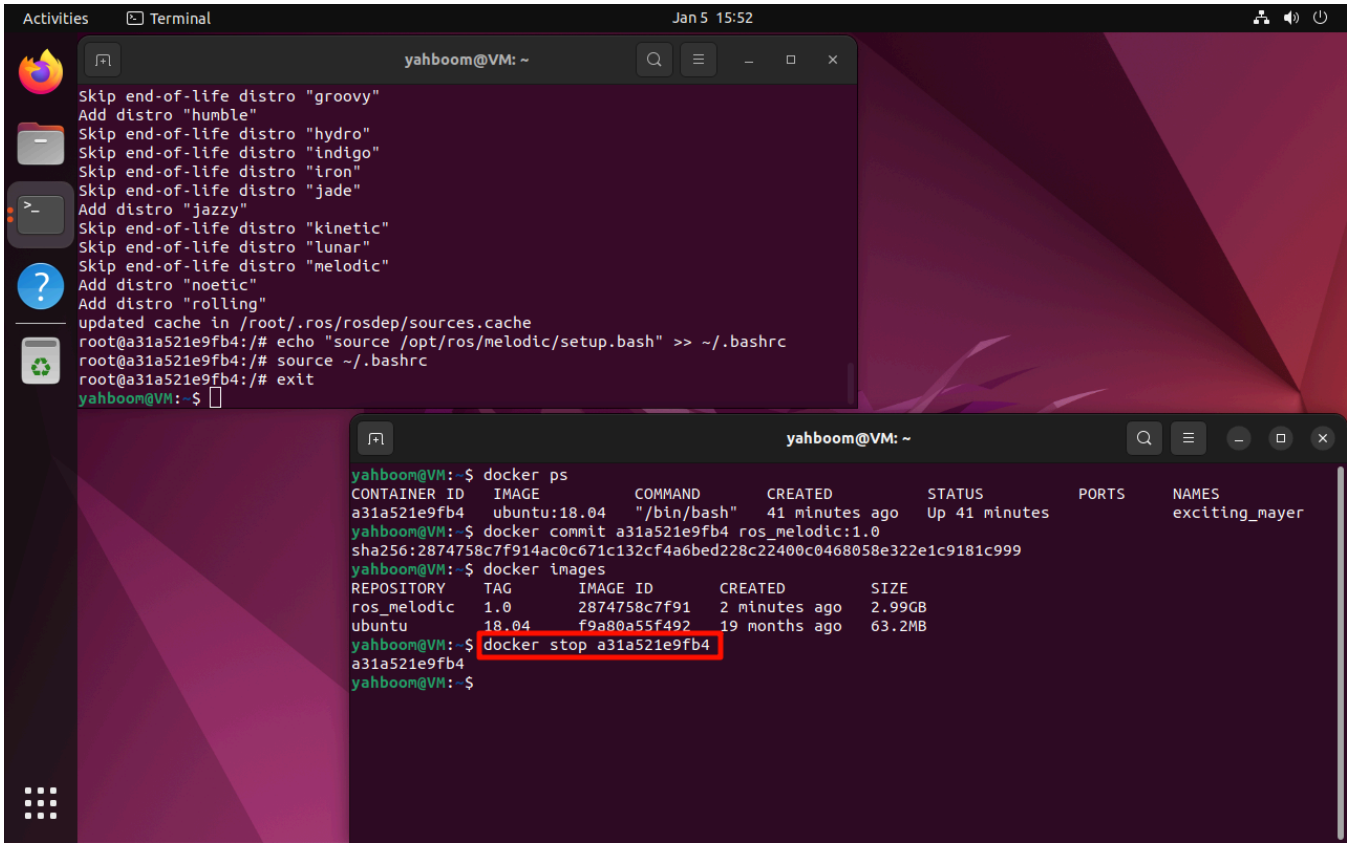
Only by committing a new image locally will the ROS environment built earlier be saved; otherwise, the next time you start the docker image, there will still be no ROS environment in the container.

Note: Modify according to the actual `CONTAINER ID`

```
docker commit a31a521e9fb4 ros_melodic:1.0
```



After saving, you can close the `ubuntu:18.04` container:



4. Verify the image/ROS

4.1. Start the image

Start the `ros_melodic:1.0` image in interactive mode: since the little turtle example has to be displayed on the graphical interface, Docker must be allowed to access the X Display

```
xhost +si:localuser:root
```

Revoke the authorization after the container exits:

```
xhost -si:localuser:root
```

```
docker run -it -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix ros_melodic:1.0
/bin/bash
```

4.2. Verify the ROS environment

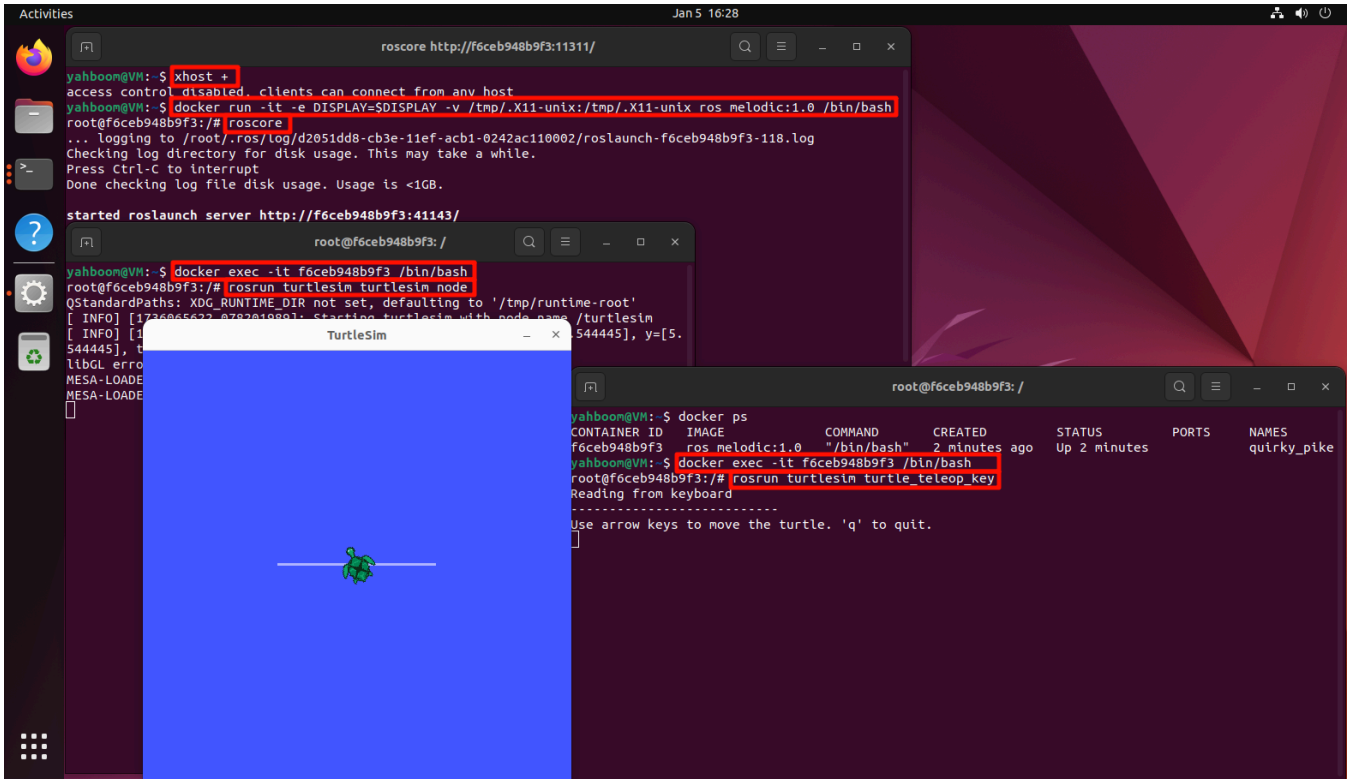
Test the ROS environment with the little turtle example: open three terminals to enter the same container, and then run the following commands respectively

```
roscore
```

```
rosrun turtlesim turtlesim_node
```

```
rosrun turtlesim turtle_teleop_key
```

Note: The mouse must stay in the third command window to control the little turtle


Activities
Jan 5 16:28
roscore http://f6ceb948b9f3:11311/
yahboom@VM:~$ xhost +
access control disabled, clients can connect from any host
yahboom@VM:~$ docker run -it -e DISPLAY=$DISPLAY -v /tmp/.X11-unix:/tmp/.X11-unix ros melodic:1.0 /bin/bash
root@f6ceb948b9f3:/# roscore
... logging to /root/.ros/log/d2051dd8-cb3e-11ef-acb1-0242ac110002/roslaunch-f6ceb948b9f3-118.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.
started roslaunch server http://f6ceb948b9f3:41143/
root@f6ceb948b9f3: /
yahboom@VM:~$ docker exec -it f6ceb948b9f3 /bin/bash
root@f6ceb948b9f3:/# rosrun turtlesim turtlesim_node
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
TurtleSim
root@f6ceb948b9f3: /
yahboom@VM:~$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
f6ceb948b9f3 ros melodic:1.0 "/bin/bash" 2 minutes ago Up 2 minutes quirky_pike
yahboom@VM:~$ docker exec -it f6ceb948b9f3 /bin/bash
root@f6ceb948b9f3:/# rosrun turtlesim turtle_teleop_key
Reading from keyboard
Use arrow keys to move the turtle. 'q' to quit.